

The Hottest Substance in the Universe

Coleridge Faraday

University of Cape Town, South Africa

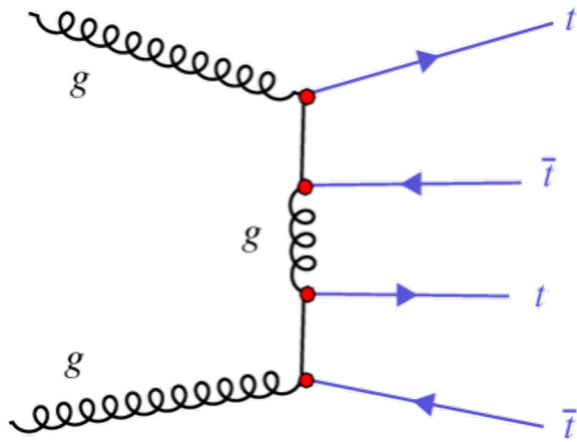
14th April 2026, NITheCS Bursary Holders Workshop

Based on

CF, B. Bert, J. Brand, W. Vogelsang, and W. A. Horowitz, [arXiv:2512.17832 \[hep-ph\]](https://arxiv.org/abs/2512.17832) (2025) and
CF and W. A. Horowitz, [JHEP **11**, 019](https://arxiv.org/abs/2501.01919) (2025)

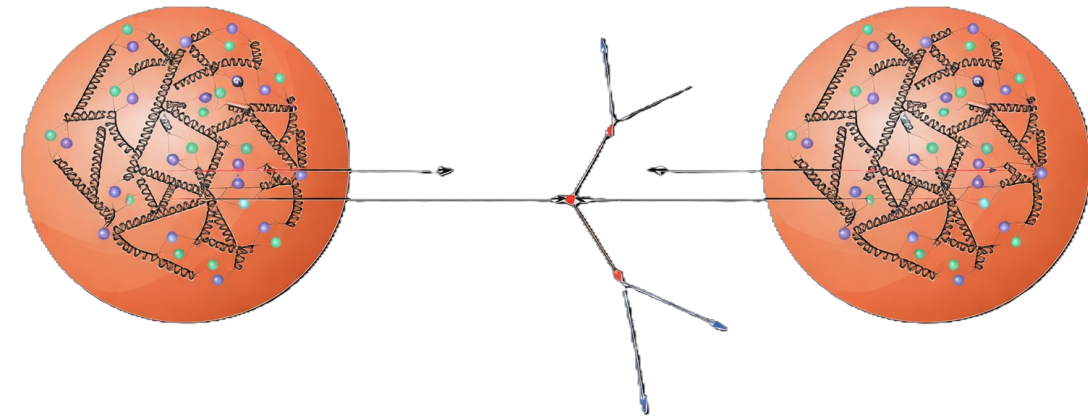


The very small

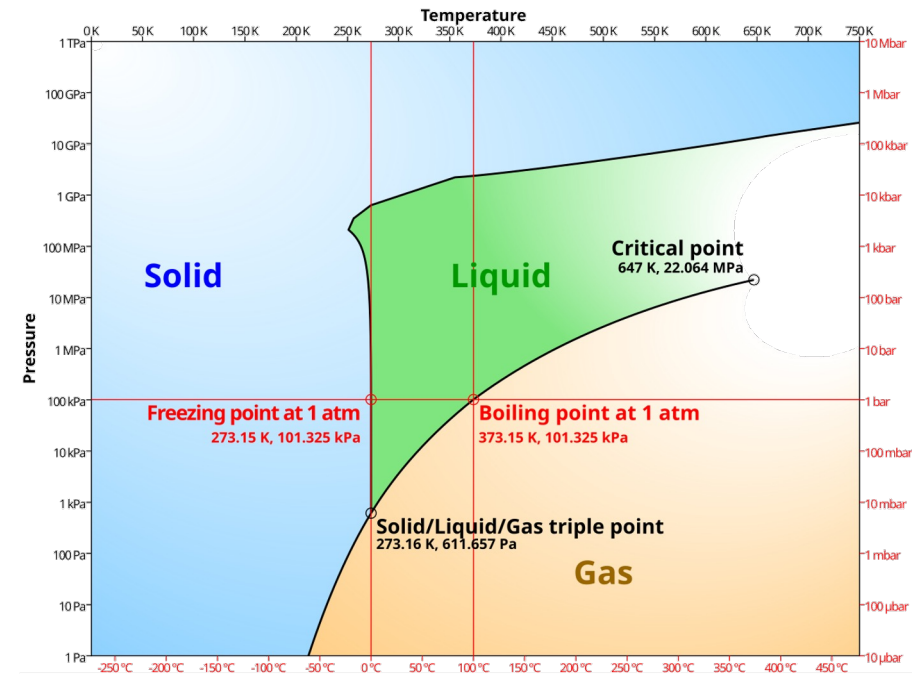
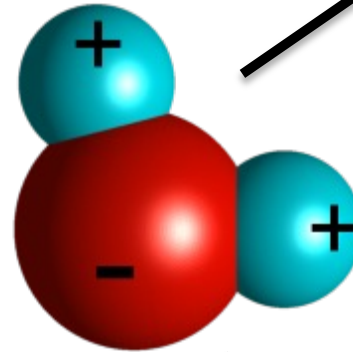


proton

proton



and



the very many

Zooming in

- Nuclei are made up of protons and neutrons
 - made up of quarks and gluons
- Under normal circumstances, quarks and gluons are confined to colour-neutral objects (like protons and neutrons) by the *strong force*

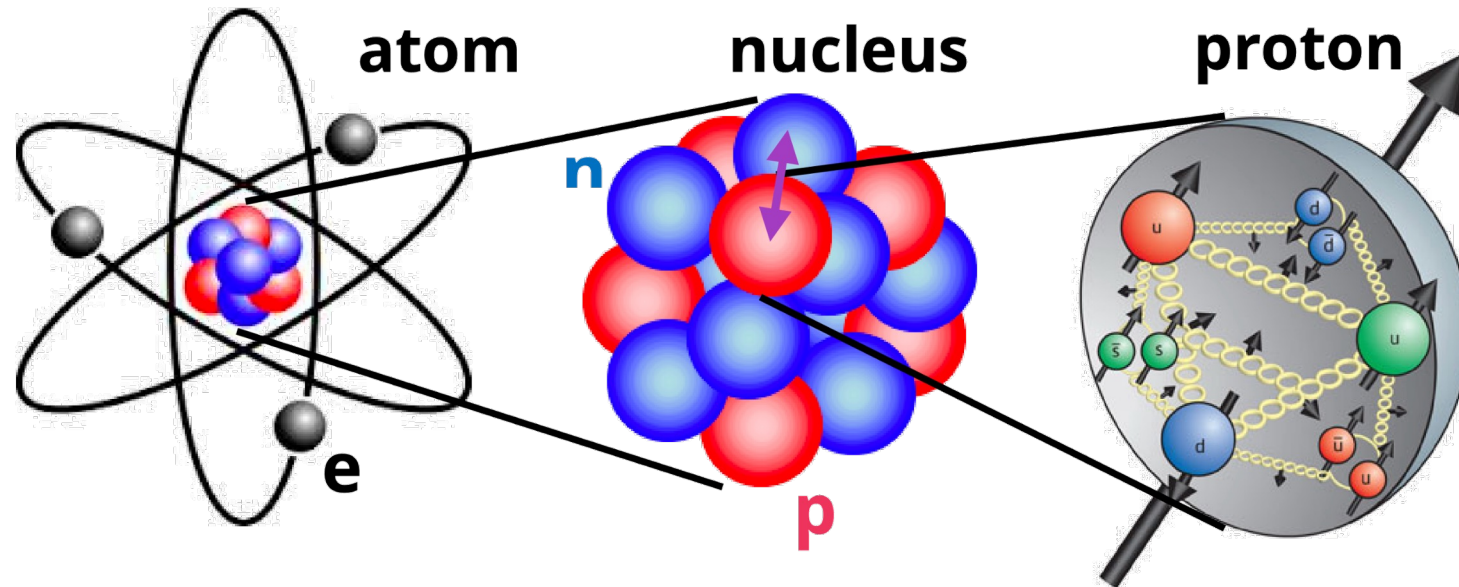
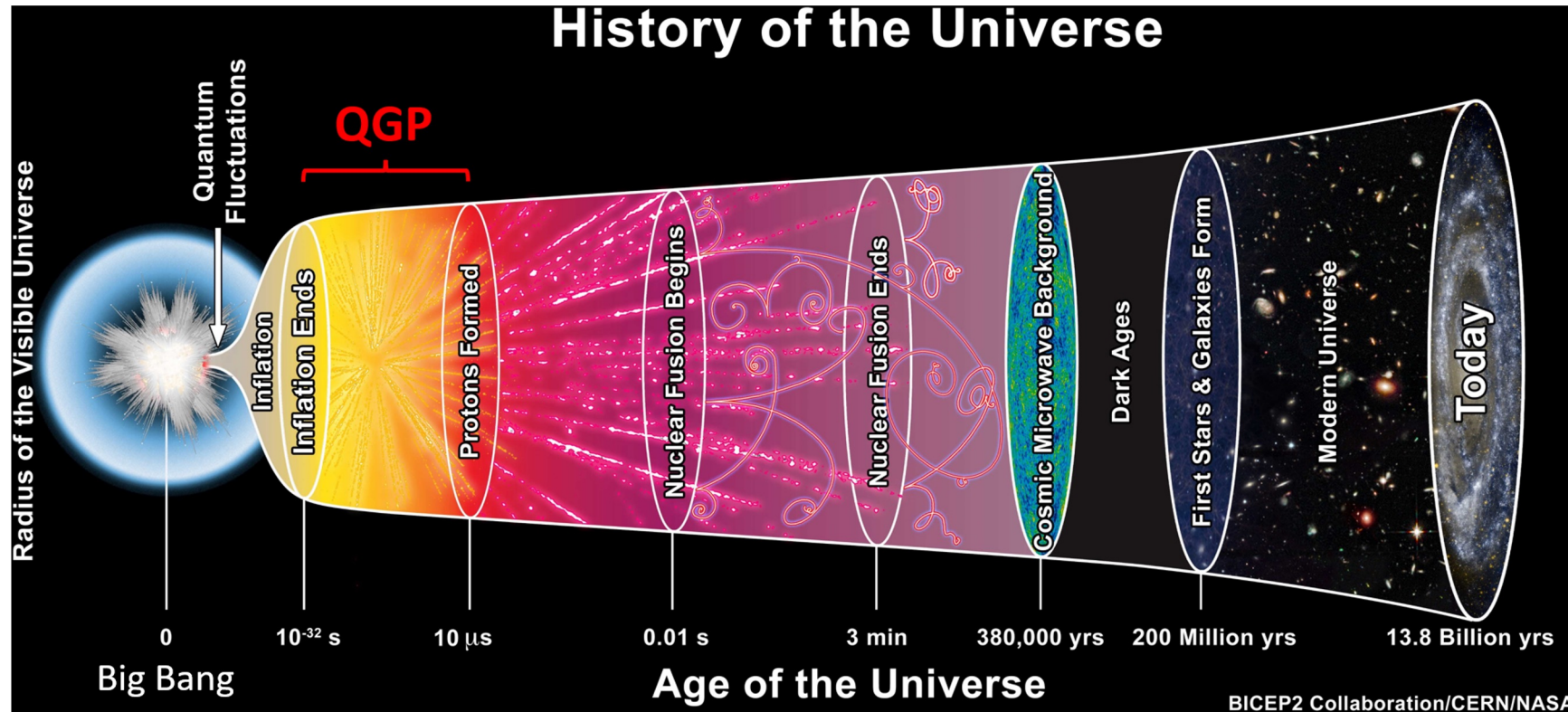


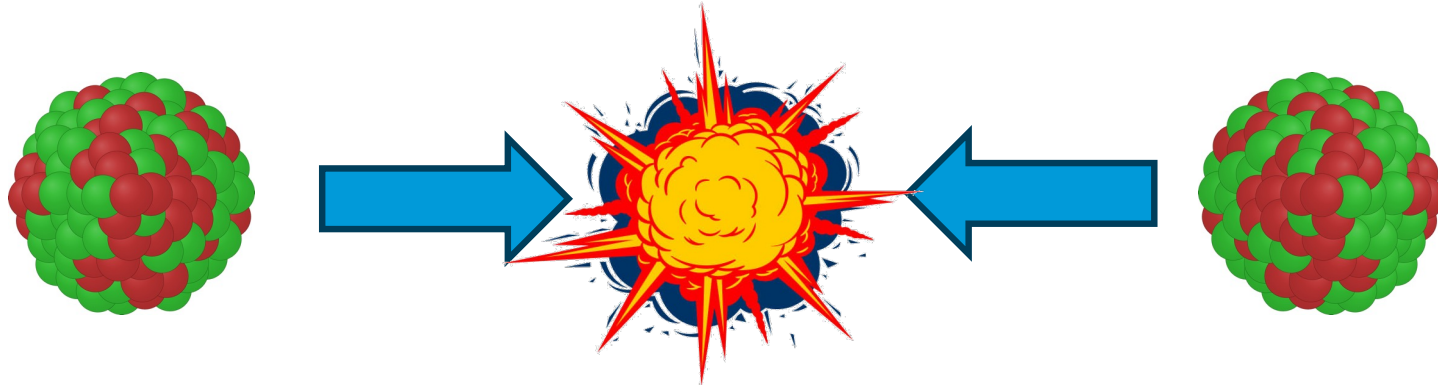
Illustration: Laura Havener

Turning up the Temperature

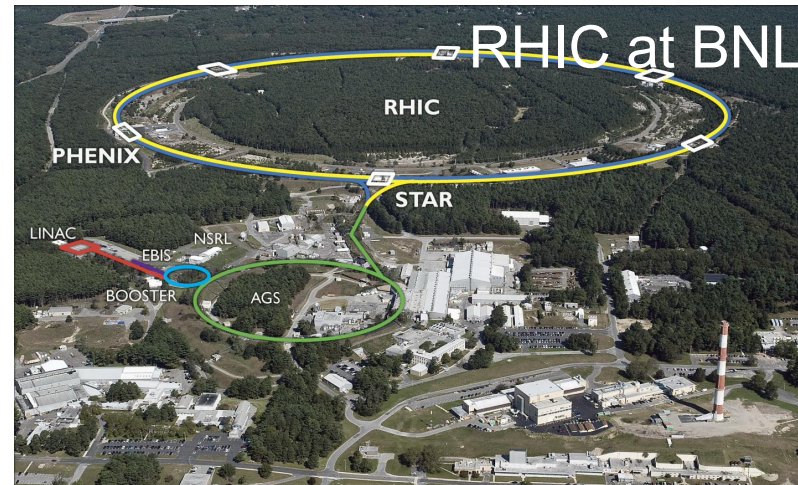
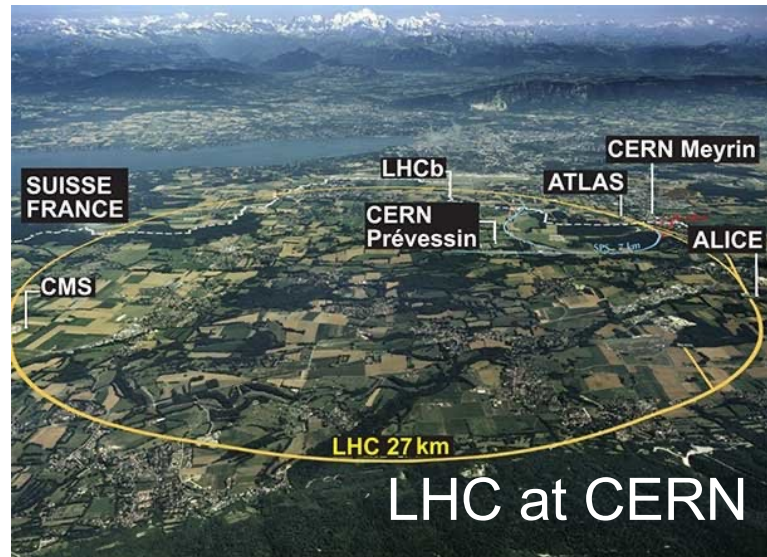
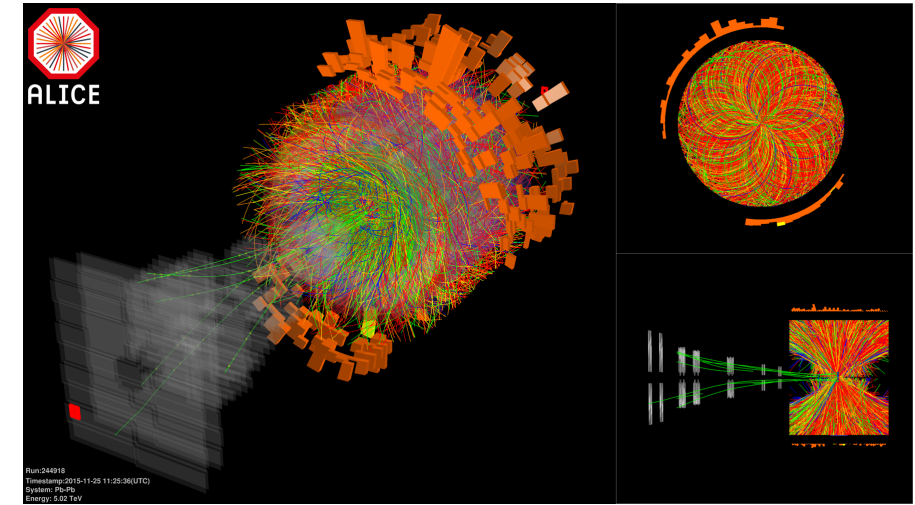


The early universe was made up of a dense, hot soup of quarks and gluons: the **quark gluon plasma**

Heavy-ion collisions

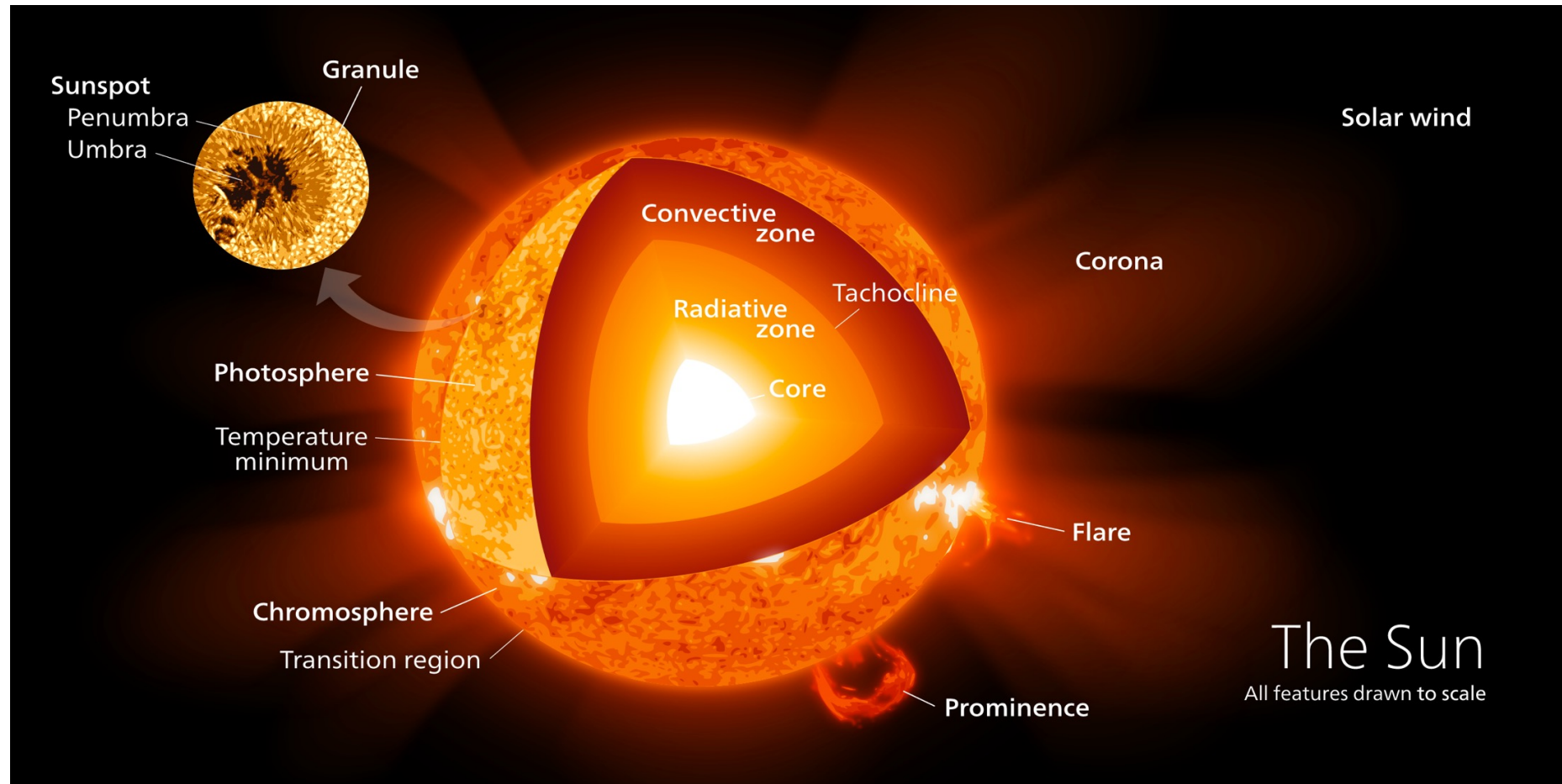


>99.9999% of the speed of light



Forms the hottest known substance in the universe, the *quark gluon plasma* at a trillion degrees!

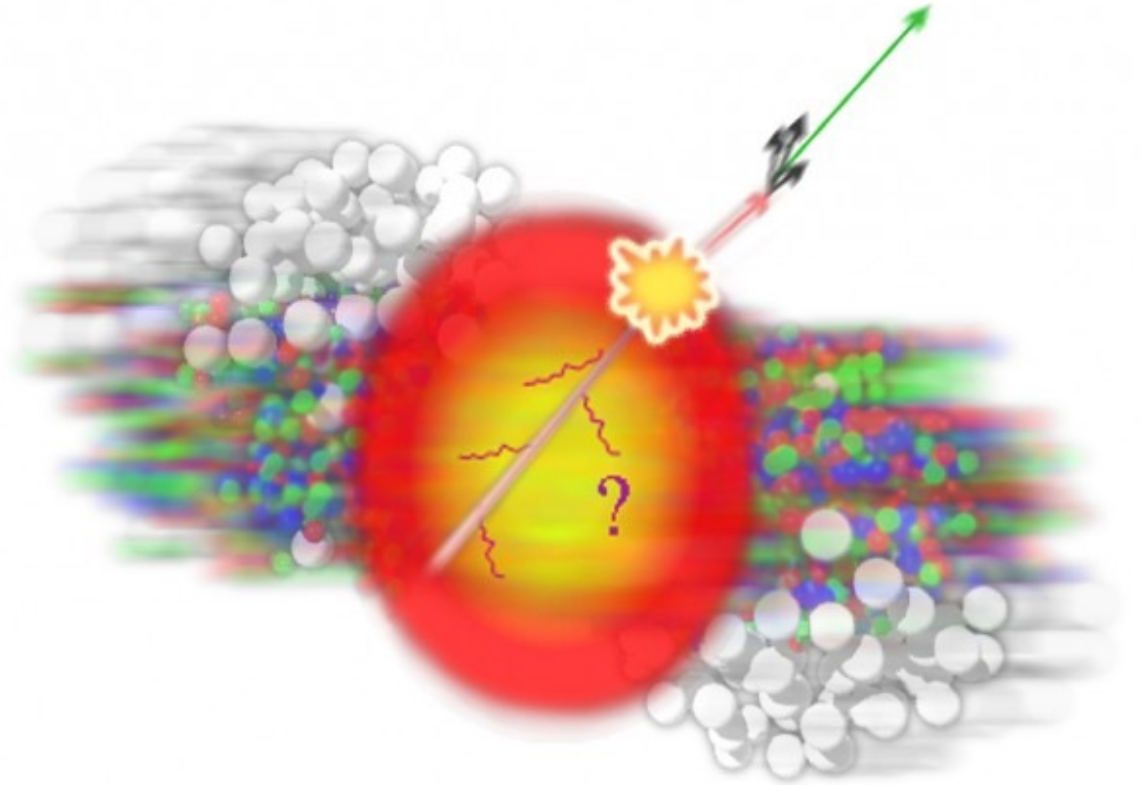
> How hot is a trillion degrees?



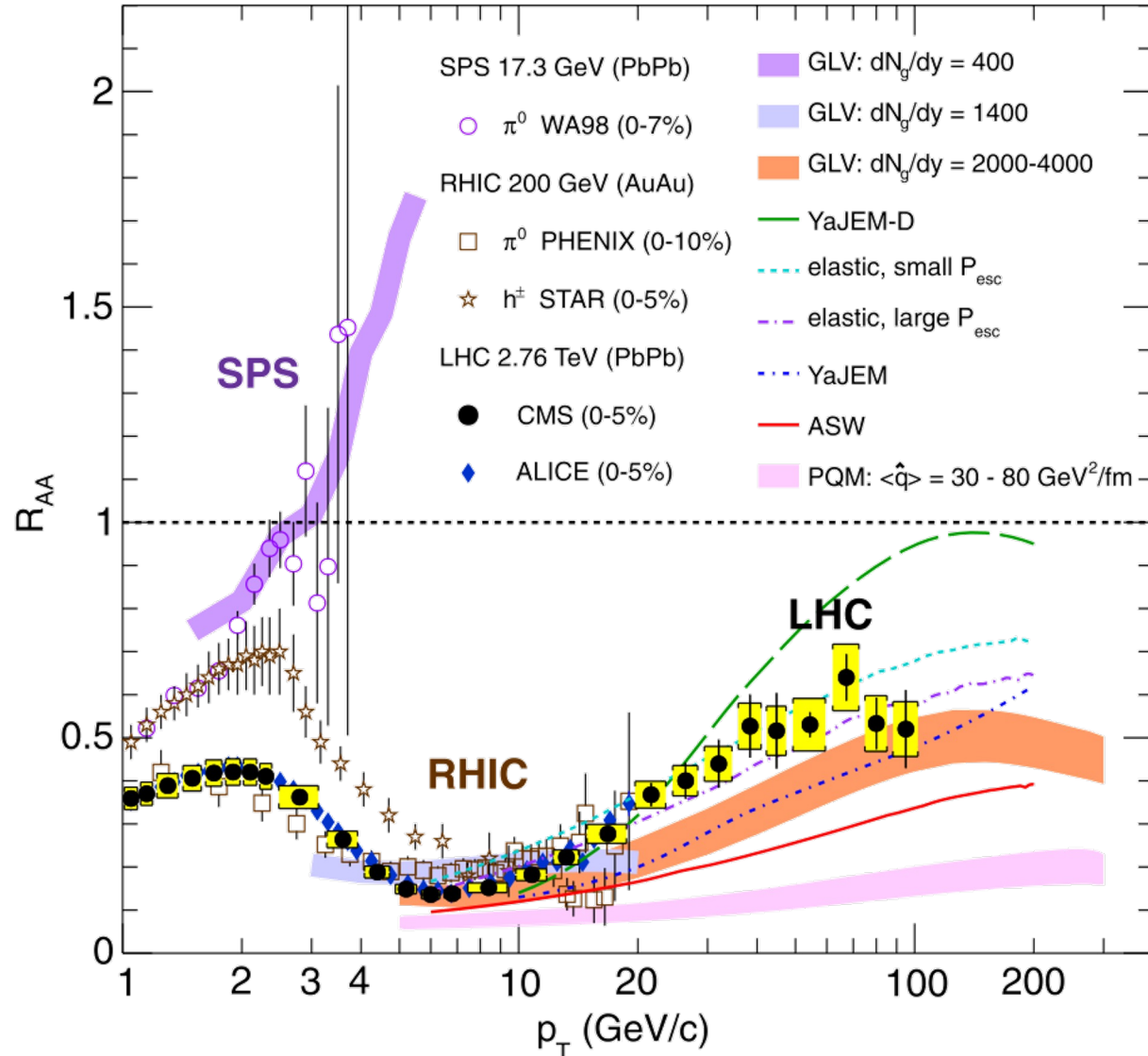
100,000x hotter than the center of the sun!

➤ How do we know a QGP is formed?

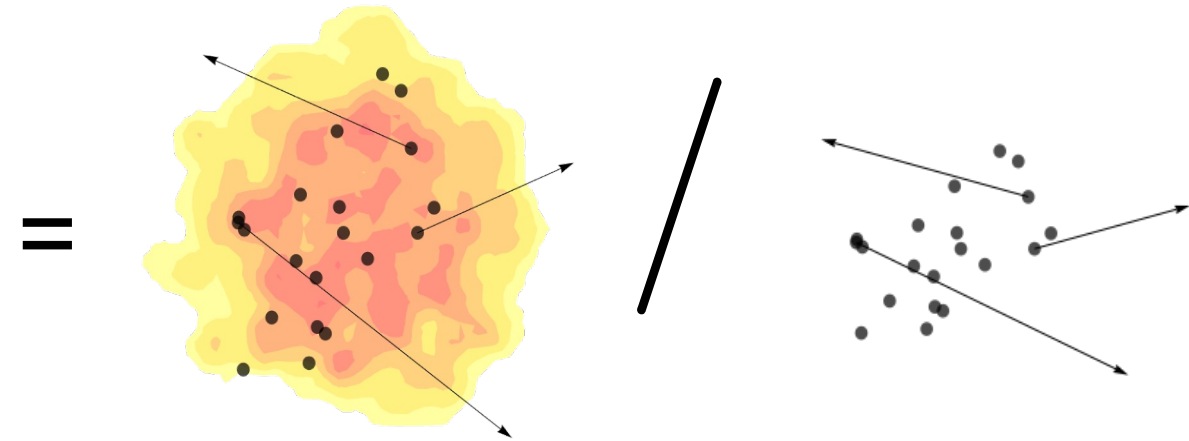
- QGP from heavy-ion collisions only lasts 10^{-23} s
- Look for *signatures* of QGP formation
- How does the QGP effect the properties (momentum, particle number, angular direction, etc.) of detected particles?



High energy probes of the QGP



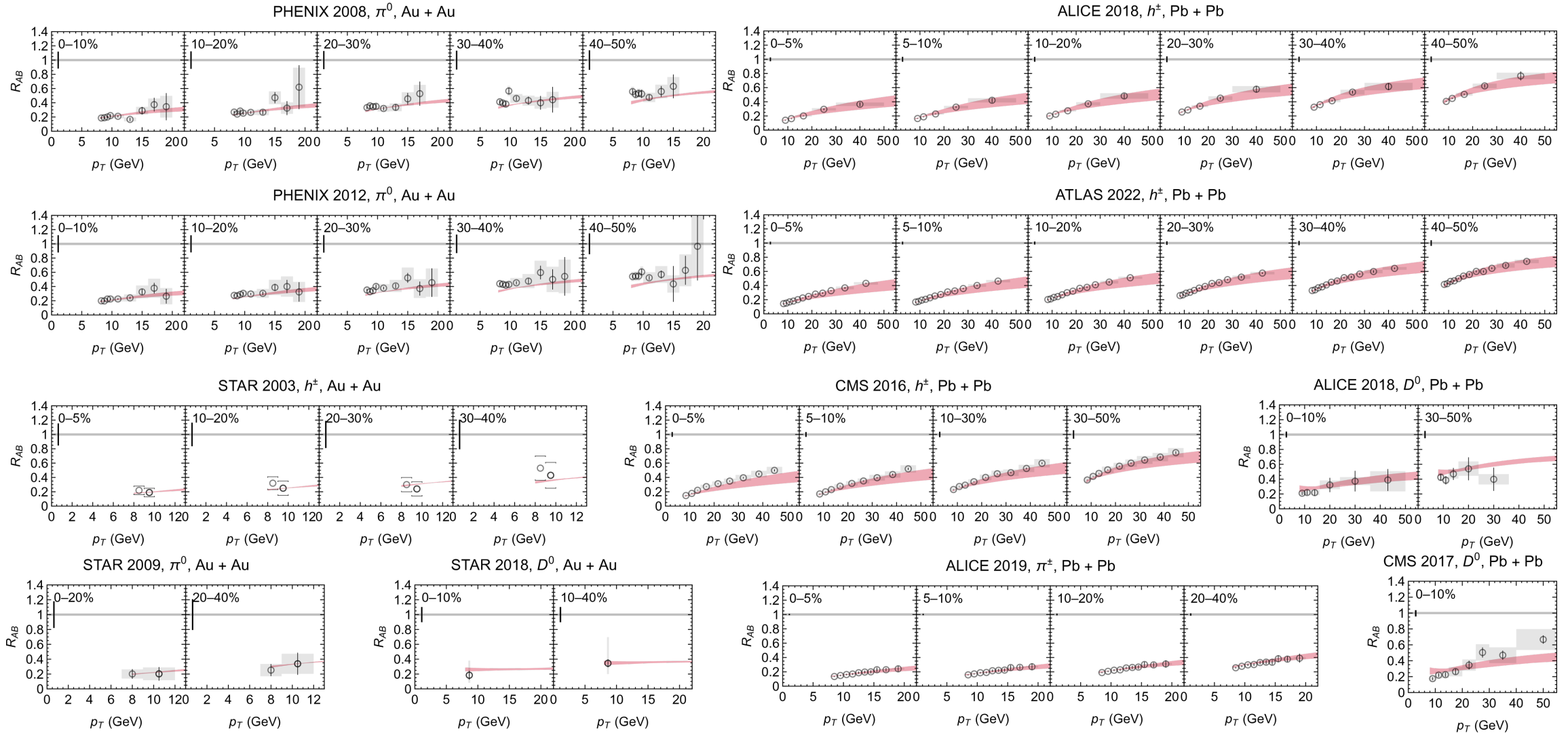
$$R_{AA}(p_T) = \frac{1}{\langle N_{coll} \rangle} \frac{dN_{AA}/dp_T}{dN_{pp}/dp_T}$$



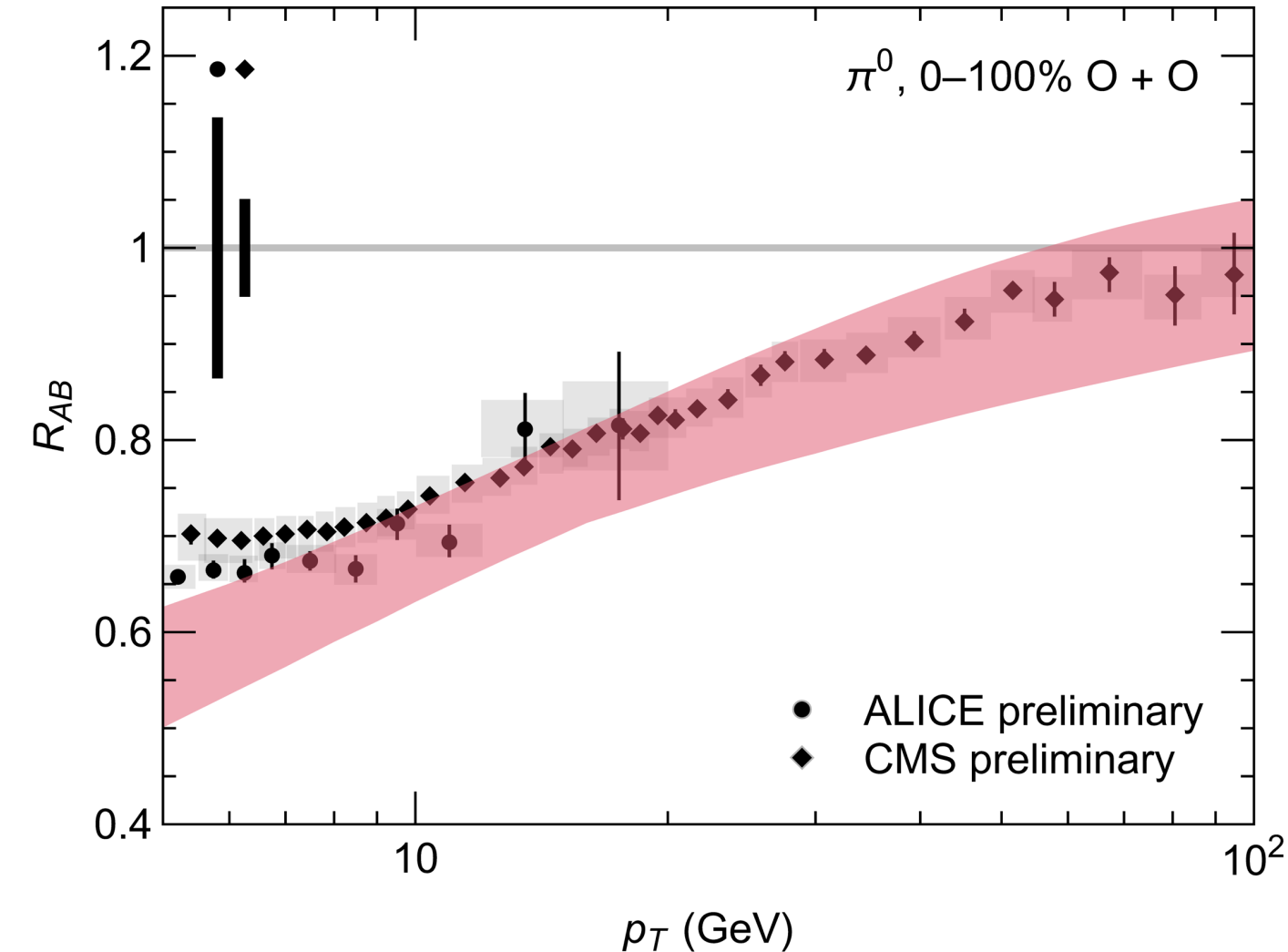
- Quantifies how much **energy is lost** by high momentum particles
- Can be used to “measure” the temperature of the QGP



Fit single parameter to data



Blind predictions for light ions

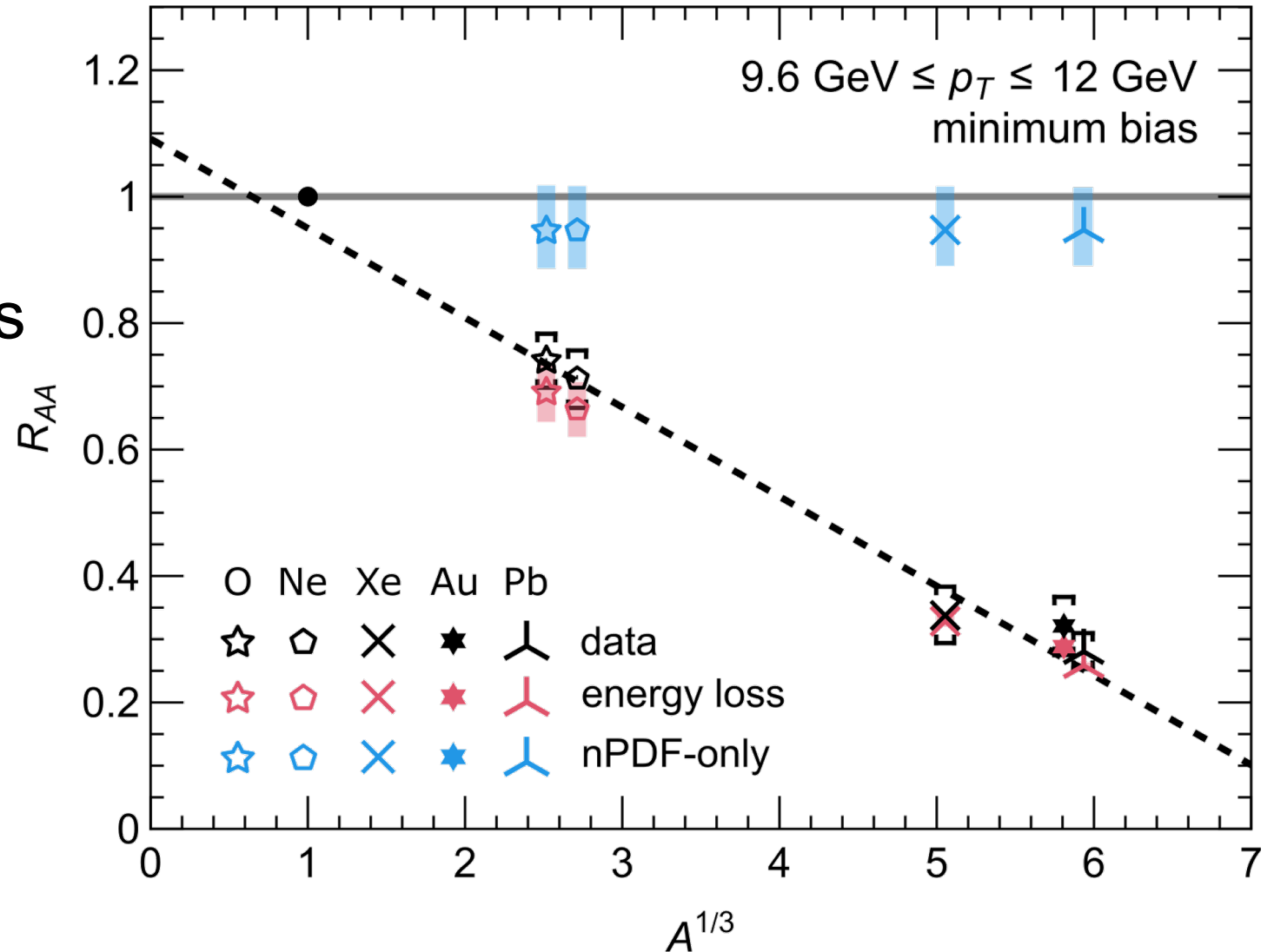


- Good agreement between our model constrained on PbPb and data

CF, W. A. Horowitz, *JHEP* 11, 019 (2025)

Results from our model

- By comparing our **model results** to **data** we can learn about the properties of the QGP in various collisions
- Colliding larger atoms leads to larger plasmas and more energy loss



CF, B. Bert, J. Brand, W. Vogelsang, and W. A. Horowitz,
arXiv:2512.17832 [hep-ph] (2025).

> What is the smallest droplet of fluid?

- Can a QGP form in extremely small collisions like proton + proton or light-ion + light-ion?
 - Many QGP signs have already been seen, even in proton-proton collisions with only ~50 particles produced!
- If so, how do the properties change?



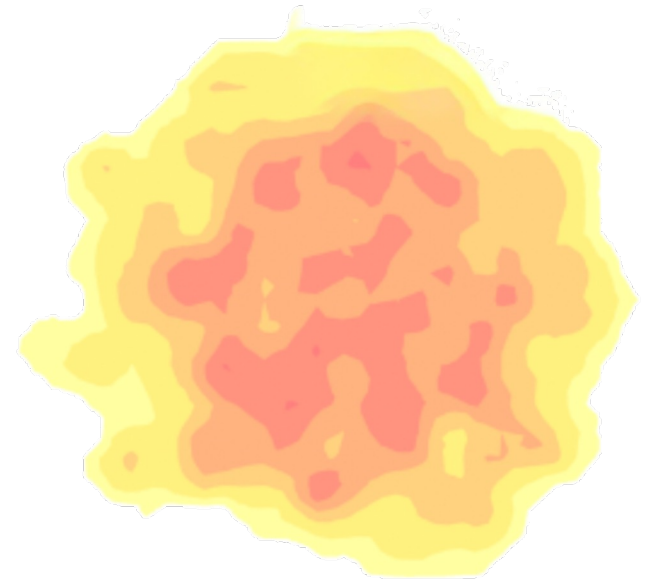
$p + p$



$p + \text{Pb}$



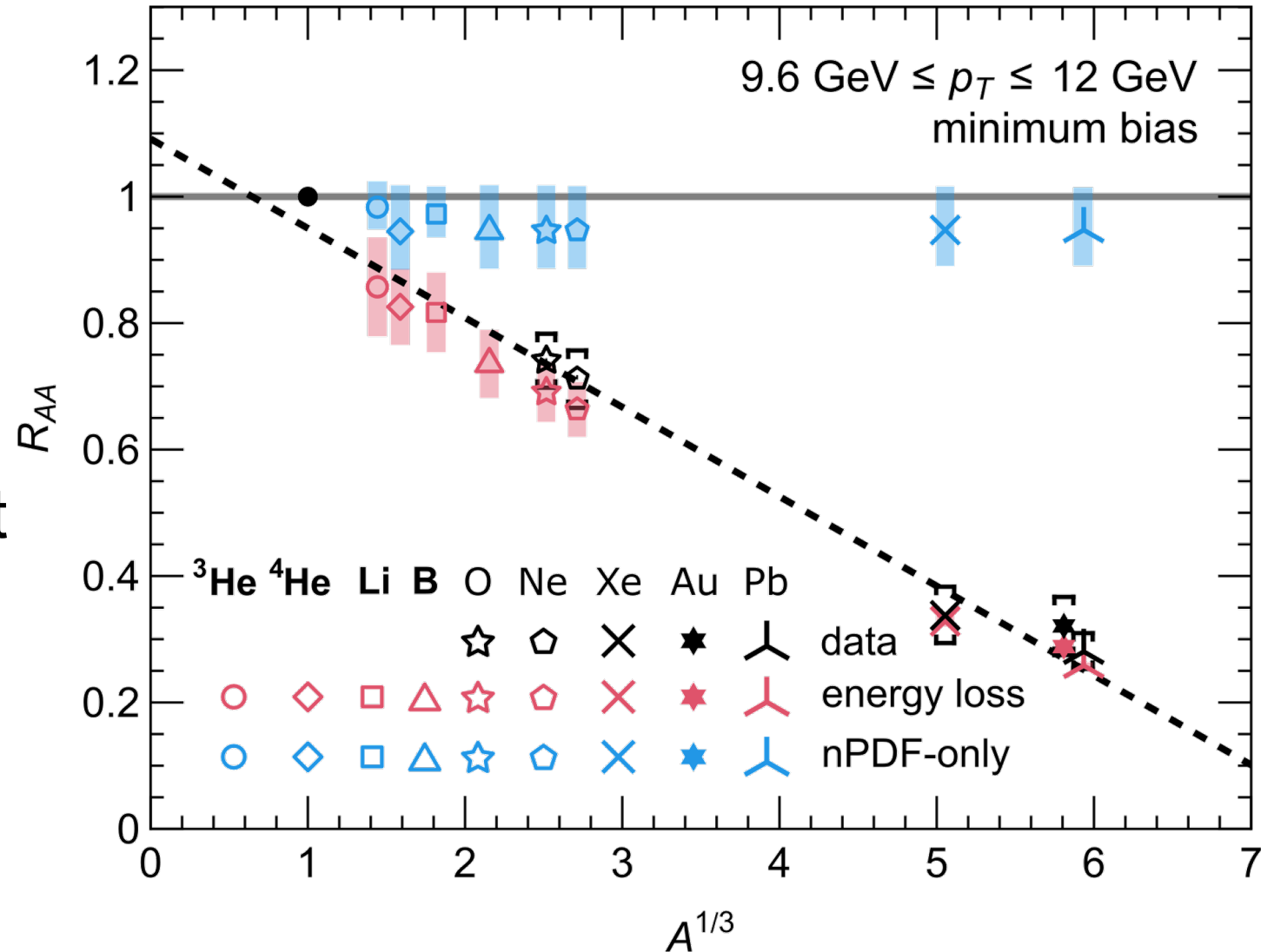
$\text{O} + \text{O}$



$\text{Pb} + \text{Pb}$

> Can we go smaller?

- We investigated numerous very light ions in our model
- These collisions could be run at the LHC
- Goal: discover the smallest droplet of QGP!



CF, B. Bert, J. Brand, W. Vogelsang, and W. A. Horowitz, arXiv:2512.17832 [hep-ph] (2025).

> Summary

- The *quark-gluon plasma* (QGP) is the state of matter formed when protons melt
- The universe was a QGP in the *first microseconds after the Big Bang*
- QGP is recreated in heavy-ion collisions on Earth
- We proposed new, lighter ions to be collided to discover the *smallest droplet of QGP*

Future (ongoing) work:

- Constrain models of the QGP to experimental data
- Study unique properties of small collision systems



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